

Base call

File

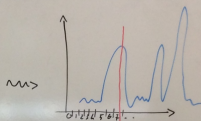
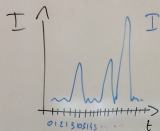
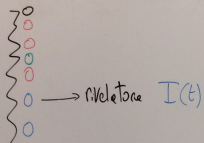
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From sensor to string (bases sequence)



Peak Detection
[7, 105, ...]

5 7 78

A C A G G C T G ...

← Sequence

base cell

$C[7, 105, 202, \dots]$

$AC[5, 28, 301, \dots]$

$G[70, 90, 160, \dots]$

$T[130, 170, 220, \dots]$

from base call to sequence

There are at least two possible approaches to go from the base call vectors to the sequence.

70	90	160						G
5	28	301						A
7	105	202						C
130	170	220						T

A C A G G C T G T C T A

from base call to sequence: cycles and lists

70	90	160						G
----	----	-----	--	--	--	--	--	---

5	28	301						A
---	----	-----	--	--	--	--	--	---

7	105	202						C
---	-----	-----	--	--	--	--	--	---

130	170	220						T
-----	-----	-----	--	--	--	--	--	---

--	--	--	--	--	--	--	--	--

from base call to sequence: cycles and lists

70	90	160						G
----	----	-----	--	--	--	--	--	---

5	28	301						A
---	----	-----	--	--	--	--	--	---

7	105	202						C
---	-----	-----	--	--	--	--	--	---

130	170	220						T
-----	-----	-----	--	--	--	--	--	---

A							
---	--	--	--	--	--	--	--

5

from base call to sequence: cycles and lists

70	90	160						G
----	----	-----	--	--	--	--	--	---

5	28	301						A
---	----	-----	--	--	--	--	--	---

7	105	202						C
---	-----	-----	--	--	--	--	--	---

130	170	220						T
-----	-----	-----	--	--	--	--	--	---

A	C						
5	7						

from base call to sequence: cycles and lists

70	90	160						G
----	----	-----	--	--	--	--	--	---

5	28	301						A
---	----	-----	--	--	--	--	--	---

7	105	202						C
---	-----	-----	--	--	--	--	--	---

130	170	220						T
-----	-----	-----	--	--	--	--	--	---

A	C	A					
5	7	28					

from base call to sequence: cycles and lists

70	90	160						G
----	----	-----	--	--	--	--	--	---

5	28	301						A
---	----	-----	--	--	--	--	--	---

7	105	202						C
---	-----	-----	--	--	--	--	--	---

130	170	220						T
-----	-----	-----	--	--	--	--	--	---

A	C	A	G				
5	7	28	70				

from base call to sequence: vectorial approach

70	90	160					
----	----	-----	--	--	--	--	--

G

5	28	301					
---	----	-----	--	--	--	--	--

A

7	105	202					
---	-----	-----	--	--	--	--	--

C

130	170	220					
-----	-----	-----	--	--	--	--	--

T

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39

from base call to sequence: vectorial approach

70	90	160					
----	----	-----	--	--	--	--	--

G

5	28	301					
---	----	-----	--	--	--	--	--

A

7	105	202					
---	-----	-----	--	--	--	--	--

C

130	170	220					
-----	-----	-----	--	--	--	--	--

T

X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39

from base call to sequence: vectorial approach

70	90	160					
----	----	-----	--	--	--	--	--

G

5	28	301					
---	----	-----	--	--	--	--	--

A

7	105	202					
---	-----	-----	--	--	--	--	--

C

130	170	220					
-----	-----	-----	--	--	--	--	--

T

X	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X	X	X	X
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

X	X	X	X	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X
20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39

from base call to sequence: vectorial approach

70	90	160						G
5	28	301						A
7	105	202						C
130	170	220						T

X	X	X	X	X	A	X	C	X	X	X	X	X	X	X	X	X	X	X	X
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

X	X	X	X	X	X	X	X	A	X	X	X	X	X	X	X	X	X	X	X
20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39

from base call to sequence: vectorial approach

70	90	160						G
5	28	301						A
7	105	202						C
130	170	220						T

A
5

C
7

A
28

from base call to sequence: vectorial approach

70	90	160						G
5	28	301						A
7	105	202						C
130	170	220						T

A	C	A
5	7	28